

Shahjalal University of Science & Technology

Department of Computer Science and Engineering
2nd Year 2nd Semester Examination, 2017

Course: CSE 233, Course Title: Object Oriented Programming
Credit: 03 Full Marks: 60 Time: 40 minutes

1. Fill In the gaps with the following words (you can use any other words if you like):
public, private, super class, sub class, main, protected, constructor, overloading, return, method, variable,
extends, argument, encapsulation, object, return type (Any six)

1*6

- i. I can have many arguments _____ variable
- ii. A method can have many of these _____ argument
- iii. I return something by definition _____ method
- iv. A class can have any number of these _____ sub class
- v. A method can have only one of these _____ return type
- vi. I act like a method but I do not have any return _____ constructor
- vii. I do not care you call me or not I will always run _____ main
- viii. Ops, we have same name but we do different things _____ overloading

2*10

1. Answer any ten questions

- i. Is constructor inherited? No
- ii. Is Empty .java file name a valid source file name? Yes
- iii. Can you make a constructor final? No
- iv. Can we execute a program without main() method? Yes
- v. What if the static modifier is removed from the signature of the main method?
will compile, but will give
NoSuchMethod error on runtime
- vi. Can you use this() and super() both in a class? Yes
- vii. Can we override the overloaded method? Yes
- viii. Can you use abstract and final both with a method? No
- ix. Can finally block be used without catch? Yes
- x. What happens if a class partially implements an interface? Abstract Class
- xi. What happens when the constructor of a class is declared as "private"?
- xii. Can a subclass variable reference a superclass object? Yes
- xiii. Can an abstract class inherit an interface? Yes

2. Answer any six questions

3*6

- i. What gives Java its 'write once and run anywhere' nature?
- ii. What purpose does the keywords throw, throws and finalize fulfill?
- iii. What is runtime polymorphism or dynamic method dispatch?

- iv. What is the meaning of immutable in terms of String?
- v. Why we cannot override static method?
- vi. How does polymorphism enable you to program "in the general" rather than "in the specific"?
- vii. What is the purpose of declaring a class, variable or method as final?
- viii. Multiple inheritance is not supported through class in java but it is possible by interface, why?

3. Answer any two questions

- i) Write a Java program to print Fibonacci series upto 100?

2

ii. Consider the following code & rectify the errors.

```

1. interface Exam{
2.     static int n;
3.     void register();
4.     default void collectTopics() {
5.         n=100;
6.     }
7.     static void completeSyllabus(){}
8. }
9. interface Syllabus extends Exam{
10. }
11. class TermTest implements Exam{
12.     public void register(){
13.         System.out.println("Study!!!");
14.     }
15.     public void main(String[] args){
16.         TermTest obj = new TermTest();
17.         Exam.collectTopics();
18.         obj.completeSyllabus();
19.     }
20. }
```

iii. What is output of the following code ?

```

1. public class ExceptionalClass {
2.     private static void sout(String prntStr) {
3.         System.out.println(prntStr);
4.     }
5.     public static void main(String[] args){
6.         try{
7.             try{
8.                 throw new NullPointerException();
9.             }catch(Exception e){
10.                 sout(e.toString());
11.                 throw new RuntimeException();
12.             }
13.         }catch(Exception e){
14.             sout(e.toString());
15.             throw new ArithmeticException();
16.         }finally{
17.             sout("Finish!");
18.         }
19.     }
20. }
```